

Unification of Matter and Forces: Bost–Connes Spectral Dynamics

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Abstract

This series of works, starting from the Bost–Connes system, has separately established arithmetic descriptions of Standard Model fermion masses [13], gauge groups and coupling constants [14], baryon spectra [16], and cosmological scales [15]. These results repeatedly use the same set of mathematical ingredients—the Hamiltonian $H = \log n$, the character group $(\mathbb{Z}/7\mathbb{Z})^*$ and its L -functions, KMS states and the $\beta = 1$ phase transition—but have never been organized as a whole. This paper unifies these ingredients into the arithmetic Dirac operator $D = H \otimes I_F + \gamma \otimes D_F$, and develops BC spectral dynamics along the logical chain: matter $\rightarrow$ forces $\rightarrow$ coupling strengths. Matter consists of the spectral eigenmodes of the Dirac operator D : the angular part corresponds to the internal quantum numbers and identity of particles, while the radial part corresponds to spatial and energy scales. Forces arise as responses of the Green kernel: angular L -functions propagate gauge forces, the radial ζ -function produces gravity, and the angular $\times$ radial combination gives rise to nuclear forces. Coupling constants are uniformly written as $\alpha_X = |A_X|^2 \eta_X$, with the distinction lying in the source of the vertex amplitude (L -function or curvature constant C^*) and the statistical treatment of internal channels (averaging or summation).

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1 Introduction

In previous works, starting from the Bost–Connes (BC) system, we separately established arithmetic descriptions of several aspects of the Standard Model:

- [13] established the correspondence between fermions and the BC system: fermion species are labeled by characters χ_k of $(\mathbb{Z}/7\mathbb{Z})^*$, and the mass formula is

$$m_f = v \times |L(\tfrac{1}{2}, \chi)|^{\text{pow}} \times (\text{nodal factor}) \times (\text{generational factor}); \quad (1)$$

- [14] established the correspondence between gauge forces and the BC system: the gauge group $U(1) \times SU(2) \times SU(3)$ is uniquely determined by the CRT decomposition, and the coupling constants of the three forces are

$$\alpha_i = \frac{|R_{k_i}|^2}{N_i}, \quad R_k = \frac{L(\tfrac{1}{2}, \chi_k)}{\zeta(\tfrac{1}{2})}; \quad (2)$$

- [15] established gravity and cosmological scales: gravity is derived from the Layer 1 Hamiltonian $H = \log n$ via the discrete Poisson equation

$$-\Delta(\log n) = R(n) = \log \frac{n^2}{n^2 - 1}, \quad (3)$$

and the unified mass scale formula is

$$E(N) = \sqrt{\frac{N \log^2 N}{C^*}}. \quad (4)$$

- [16] established the correspondence between baryons and the BC system: baryons appear as collective emergent states at the ζ pole, with masses

$$E_n = \sqrt{\frac{\zeta(\beta) \log^2 \zeta(\beta)}{C^*}}, \quad \beta = 1 + \frac{1}{\sqrt{n}}, \quad (5)$$

and internal quantum numbers are transmitted by characters through the color singlet condition, reflecting the bipartite structure of the Dirac operator;

These works are not unrelated formulas. They repeatedly use the same set of mathematical objects: the BC system Hamiltonian $H|n\rangle = (\log n)|n\rangle$, characters χ_k of $(\mathbb{Z}/7\mathbb{Z})^*$ and their L -functions, KMS states and the $\beta = 1$ phase transition, and the CRT decomposition $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. The way these objects relate to each other—who describes particles, who propagates forces, who determines coupling strengths—appears in different aspects in each paper, but has never been stated as a whole.

The purpose of this paper is to unify this common set of objects into **BC spectral dynamics**. It consists of three sets of data, each answering a fundamental physical question:

1. **What particles exist?**—*Spectral content*. The eigenvalues and eigenspaces of the arithmetic Dirac operator $D = H \otimes I_F + \gamma \otimes D_F$ determine the species, quantum numbers, and masses of particles. The angular part corresponds to particle identity (species, internal quantum numbers), and the radial part corresponds to spatial and energy scales.

2. **How do particles interact?**—*Green response*. The KMS kernel $K_0(\beta) = \zeta(\beta)$ and the Dirichlet kernel $K_\chi(\beta) = L(\beta, \chi)$ are the linear responses of the system to external sources. Angular Green kernels propagate gauge forces (electromagnetic, weak, strong), the radial Green kernel $\zeta(\beta)$ propagates gravity, and the normalized response $R_\chi = L/\zeta$ connects matter and forces.
3. **How strong are the forces?**—*Born response trace*. The coupling strength $\alpha_X = |A_X|^2 \eta_X$ is jointly determined by the amplitude A_X and channel statistics η_X , following Born's probability rule.

This paper is organized along the logical chain of these three sets of data. §2 establishes the common spectral data (Dirac operator, Green kernel, and vertex amplitude), §3 identifies matter as eigenmodes of the spectrum (fermion identity, baryon emergence, and mass origin), §4 identifies interactions as Green responses (gauge group, discrete gauge equations, and gravity), §5 gives the unified response trace formula for coupling constants, §6 demonstrates spectral completeness.

2 Arithmetic Structure of Spectral Dynamics

In p -adic number theory, every nonzero p -adic integer uniquely decomposes into the product of a unit part and a power part. At $p = 7$, this decomposition is

$$\mathbb{Z}_7 \setminus \{0\} \cong \mathbb{Z}_7^* \times 7^{\mathbb{N}}, \quad (6)$$

i.e., every $x \in \mathbb{Z}_7 \setminus \{0\}$ is uniquely written as $x = u \cdot 7^r$, where $u \in \mathbb{Z}_7^*$ (unit) and $r \in \mathbb{N}$ (valuation). This is the defining property of the 7-adic valuation and involves no choices.

This decomposition separates the degrees of freedom of the BC system into two independent directions.

Angular: the unit group $\mathbb{Z}_7^*$. The angular direction controls the internal quantum numbers of particles. At the first approximation level (modulo 7), the unit group is the finite group

$$(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}. \quad (7)$$

The last step is the Chinese Remainder Theorem (CRT) decomposition, where $\mathbb{Z}/2\mathbb{Z}$ corresponds to the weak isospin direction and $\mathbb{Z}/3\mathbb{Z}$ corresponds to the color direction. The bipartite (weak) and tripartite (color) structure required by the Standard Model gauge group $U(1) \times SU(2) \times SU(3)$ is precisely provided by $\varphi(7) = 6 = 2 \times 3$. This decomposition is unique: the prime factors of 6 are exactly 2 and 3, with no third independent direction.

Radial: the valuation layer 7^r . The radial direction controls the energy scale. The spectrum $\{\log n : n \in \mathbb{N}^*\}$ of the Hamiltonian $H = \log n$ naturally measures radial distance, and the partition function $\text{Tr } e^{-\beta H} = \zeta(\beta)$ controls state density and phase transitions.

Independence. The spectra of the two directions are independent: angular characters depend only on u , not on r ; radial energy depends only on r , not on u . This independence allows the subsequent sections to separately analyze the angular (fermions and gauge forces), radial (gravity), and angular×radial combined (baryons and nuclear forces) aspects, then combine them through the arithmetic Dirac operator $D = H \otimes I_F + \gamma \otimes D_F$.

Proposition 2.1 (Angular/radial independence). *Let χ_k be a character of $(\mathbb{Z}/7\mathbb{Z})^*$, and $H|n\rangle = (\log n)|n\rangle$ the radial Hamiltonian. Then:*

- (i) *Radial energy does not distinguish character directions: $\langle \chi_k | H | \chi_k \rangle = \langle \chi_0 | H | \chi_0 \rangle$ holds for all k ;*
- (ii) *Angular eigenvalues $d_k = \sum_a c_a \chi_k(a)$ do not depend on n .*

Proof. (i) $H|n\rangle = (\log n)|n\rangle$ is uniform for all n and does not depend on the character label. (ii) d_k involves only summation over the finite group $(\mathbb{Z}/7\mathbb{Z})^*$ and contains no positive integer n . $\square$

2.1 Angular Structure: D_F and Δ_{mult}

The angular degrees of freedom are carried by the group algebra $\mathbb{C}[(\mathbb{Z}/7\mathbb{Z})^*]$ of the finite group $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}$. The CRT decomposition (7) has already separated it into the $\mathbb{Z}/2\mathbb{Z}$ (weak) and $\mathbb{Z}/3\mathbb{Z}$ (color) directions. Below we construct the operators acting on this space.

Additive shift operator T_a . T_a ($a \in (\mathbb{Z}/7\mathbb{Z})^*$) is defined as

$$(T_a f)(g) = f(g + a), \quad g \in (\mathbb{Z}/7\mathbb{Z})^*. \quad (8)$$

T_a shifts functions on the multiplicative group $(\mathbb{Z}/7\mathbb{Z})^*$ along the *additive* direction—this operation mixes the additive structure and multiplicative structure of $(\mathbb{Z}/7\mathbb{Z})$, and is the carrier of the “additive–multiplicative tension” in the BC system. Physically, T_a connects different prime channels: additive shifts change the identity of multiplicative group elements, producing couplings between particles.

Finite Dirac operator D_F . D_F is defined as a linear combination of additive shifts [3]:

$$D_F = \sum_{a \in (\mathbb{Z}/7\mathbb{Z})^*} c_a T_a, \quad (9)$$

where the coefficients c_a are uniquely determined by the axioms of the spectral triple (self-adjointness, compact resolvent, regularity) [3, 8].

D_F is diagonalized in the character basis $\{\chi_k\}$. Character χ_k is an eigenfunction of D_F :

$$D_F \chi_k = d_k \chi_k, \quad d_k = \sum_{a \in (\mathbb{Z}/7\mathbb{Z})^*} c_a \chi_k(a). \quad (10)$$

The eigenvalue d_k encodes the effective strength of additive shifts in the χ_k direction. d_k quantifies the degree of additive–multiplicative non-commutativity in the χ_k direction: $d_k = 0$ means that addition and multiplication completely commute in that direction; $d_k \neq 0$ means that addition and multiplication do not commute. The larger d_k is, the stronger the tension, and the richer the physics carried by that direction.

Additive–multiplicative commutator $[D_F, \mu_g]$. The commutator of D_F with the multiplicative shift μ_g (a generator of the BC system) measures the non-commutativity of addition and multiplication:

$$[D_F, \mu_g] \chi_k = d_k (\chi_k(g) - 1) \mu_g \chi_k. \quad (11)$$

When $\chi_k(g) \neq 1$, the commutator is nonzero—addition and multiplication do not commute in the χ_k direction; when $\chi_0(g) \equiv 1$, the commutator is identically zero—addition and multiplication completely commute in the vacuum direction.

This commutator is the root of gauge forces in the BC system. The nonvanishing of the commutator is the condition for the existence and propagation of gauge forces. The magnitude $|d_k(\chi_k(g) - 1)|$ of the commutator directly controls the propagation strength of the gauge force in that direction.

Angular multiplicative Laplacian Δ_{mult} . The angular multiplicative Laplacian is defined as

$$\Delta_{\text{mult}} = S + S^{-1} - 2I, \quad (12)$$

where S is the shift operator of the generator $g = 3$ of $(\mathbb{Z}/7\mathbb{Z})^*$ ($Sf(x) = f(gx)$, multiplicative shift). Δ_{mult} is diagonalized in the character basis, with eigenvalues

$$\lambda_k = 2 - 2\cos(2\pi k/6), \quad k = 0, 1, \dots, 5, \quad (13)$$

taking four distinct values $\{0, 1, 3, 4\}$. λ_k quantifies how much the multiplicative shift deviates from the identity—the direction with $\lambda = 0$ is completely stationary (center/Cartan), while directions with $\lambda \neq 0$ oscillate under multiplicative shifts. The eigenvalues of Δ_{mult} label the sector classification of gauge fields.

Relationship between D_F and Δ_{mult} . D_F (§2.1) is a linear combination of additive shifts T_a ; Δ_{mult} is the second-order difference of the multiplicative shift $S = \mu_g$. Both act on the same space $\mathbb{C}[(\mathbb{Z}/7\mathbb{Z})^*]$ and are both diagonalized in the character basis, but encode different structures:

	D_F	Δ_{mult}
Construction	Linear comb. of additive shifts T_a	2nd-order diff. of mult. shift S
Eigenvalues	$d_k = \sum_a c_a \chi_k(a)$	$\lambda_k = 2 - 2\cos(2\pi k/6)$
Encodes	Add.-mult. tension (matter)	Mult. shift oscillation (forces)
Physical role	Matter	Forces

Both are derived from the two algebraic operations (addition vs. multiplication) of the same finite group $(\mathbb{Z}/7\mathbb{Z})^*$, describing matter and forces respectively.

2.1.1 Physical Meaning of the Angular Direction

The angular structure describes three types of physical information: particle identity, force classification, and force propagation. Each is explained below.

Particle identity. The identity of each particle is determined by its **angular position**—i.e., its direction in character space. The CRT decomposition $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ provides two independent quantum number directions: weak isospin ($\mathbb{Z}/2\mathbb{Z}$) and color ($\mathbb{Z}/3\mathbb{Z}$). The eigenvalue d_k of D_F labels the additive–multiplicative tension strength of each direction. χ_0 is the vacuum reference direction; baryons project back to the χ_0 direction as color singlets via ε^{abc} (quantum numbers are summed from constituents). The normalized response $|R_k|^2 = |L(1/2, \chi_k)/\zeta(1/2)|^2$ gives the path weight of each fermion species in the energy scale—different particles take different character paths, L -function values differ, and mass ratios arise accordingly.

Identity	character	ord	d_k	CRT	weak	color	Q
vacuum	χ_0	1	0	—	—	—	—
neutrino	χ_3	2	4	$(1, 0)$	doublet	singlet	0
charged lepton	χ_2, χ_4	3	3	$(0, \pm 1)$	singlet	singlet	-1
quark	χ_1, χ_5	6	1	$(1, \pm 1)$	doublet	triplet	$2/3, -1/3$
baryon	color singlet projection	—	—	color singlet combination	inherited	singlet	+1, 0

Force classification. The eigenvalues of Δ_{mult} on the adjoint representation (endomorphism space $\text{End}(\mathbb{C}[(\mathbb{Z}/7\mathbb{Z})^*])$) classify gauge fields into four sectors according to CRT coordinate differences. Let $g_1, g_2 \in (\mathbb{Z}/7\mathbb{Z})^*$; the gauge field endomorphism $|g_1\rangle\langle g_2|$ is labeled by the CRT coordinate difference $(a, b) \in \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ of $g_1 - g_2$. The principle of CRT-factor-preserving automorphisms excludes the $\lambda = 1$ sector that simultaneously changes both the $\mathbb{C}^2$ and $\mathbb{C}^3$ factors—such endomorphisms do not preserve the factor algebras $\mathcal{A}_2$ or $\mathcal{A}_3$. Weak–color mixing is forbidden at all energy scales. The remaining three sectors correspond exactly to the three gauge forces of the Standard Model:

CRT coord. diff. (a, b)	Adjoint character	λ	Force	Status
$(0, 0)$	χ_0	0	Electromagnetic ($U(1)$ center/Cartan)	Allowed
$(0, \pm 1)$	$\chi_2 \oplus \chi_4$	3	Color force (gluons)	Allowed
$(1, 0)$	χ_3	4	Weak force (W/Z)	Allowed
$(1, \pm 1)$	$\chi_1 \oplus \chi_5$	1	Weak–color mixing	Forbidden

Force propagation. The nonvanishing of the commutator $[D_F, \mu_g]$ is the root of the existence and propagation of gauge forces. In the χ_0 direction the commutator vanishes—addition and multiplication completely commute, producing no gauge force. Gravity does not propagate through the angular direction; it is a structural property of the radial n -space (§2.2).

In the χ_k ($k \neq 0$) directions the commutator is nonzero—additive–multiplicative tension produces gauge force propagation. The angular Green kernel is the Dirichlet L -function $K_\chi(\beta) = L(\beta, \chi_k)$, and the physically observable propagation amplitude is the normalized response $R_k(\beta) = L(\beta, \chi_k)/\zeta(\beta)$. β is the Mellin dual variable of the radial n -space (momentum representation), β is the Mellin dual variable (momentum representation) of the radial n -space; the propagation equation is most concise in β -space.

Direction	$[D_F, \mu_g]$	Green kernel	Propagator	Physics
$\chi_0(\text{vacuum})$	$= 0$	$\zeta(\beta)$	—	No angular propagation; gravity is radial
$\chi_k(k \neq 0)$	$\neq 0$	$L(\beta, \chi_k)$	$R_k = L/\zeta$	Gauge force propagation

2.2 Radial Structure: Hamiltonian $H = \log n$

The radial degree of freedom is governed by the Hamiltonian H , acting on the positive-integer Hilbert space $\ell^2(\mathbb{N}^*)$:

$$H|n\rangle = (\log n)|n\rangle, \quad n \in \mathbb{N}^*. \quad (14)$$

The spectrum of H is $\{\log n : n \in \mathbb{N}^*\}$. Each prime p contributes an eigenvalue $\log p$, serving as the most fundamental arithmetic excitation; the energy of a composite number $n = p_1^{a_1} \cdots p_k^{a_k}$ is $\sum_i a_i \log p_i$, reflecting the additive structure of the unique factorization theorem.

The partition function of H directly yields the Riemann ζ function:

$$\text{Tr } e^{-\beta H} = \sum_{n \geq 1} n^{-\beta} = \zeta(\beta). \quad (15)$$

$\zeta(\beta)$ has a unique simple pole at $\beta = 1$, corresponding to a divergence in the density of states. This pole marks the KMS phase transition of the BC system from the disordered phase to the ordered phase, and physically corresponds to the Higgs mechanism (§3.2) and baryon emergence (§4.3).

The radial structure describes all physics related to space and energy: physical distance, spatial curvature (gravity), and energy scales.

Physical space. The positive integer n serves simultaneously as a state-space label and as the radial distance coordinate of physical space. $H = \log \hat{N}$ is a function of $\hat{N}$; the two commute and share the eigenbasis $|n\rangle$ —“the particle is in a state of energy $\log n$ ” and “the particle is at position n ” are described by the same basis vector. The additive metric $d_+(m, n) = |m - n|$ defines physical distance.

Spatial curvature (gravity). Gravity is not a force propagating in space, but the curvature of space itself. The discrete Poisson equation

$$-\Delta(\log n) = R(n) = \log \frac{n^2}{n^2 - 1} \sim n^{-2} \quad (16)$$

is defined directly on $\mathbb{N}^*$. The gravitational equation naturally carries n , because gravity is a structural property of n -space. The discrete curvature constant

$$C^* = 2 \sum_{n \geq 1} \log^2(1 + 1/n) = 1.954378 \quad (17)$$

enters the computation of all radial physical quantities.

Energy scales. The unified scale formula

$$E(N) = \sqrt{\frac{N \log^2 N}{C^*}} \quad (18)$$

yields the three hierarchies of the Standard Model when N takes different values:

N	Physical scale
$\zeta(1 + 1/\sqrt{d_2})$	m_p
$\varphi(p^{d_2+1})$	v_{EW}
$\varphi(7^{K_{Pl}})$	M_{Pl}

All three scales arise from the same formula, the same constant C^* , and different values of N . The choice of N is determined by the KMS structure of the BC system: the hot phase ($\beta \rightarrow 1^+$) reads the partition function $\zeta(\beta)$, and the cold phase ($\beta \gg 1$) reads the algebraic mode count $\varphi(p^K)$.

KMS phase transition. The unique pole of $\zeta(\beta)$ at $\beta = 1$ marks the ordered→disordered KMS phase transition. For $\beta > 1$ the KMS state is non-unique (spontaneous symmetry breaking); for $\beta \leq 1$ the KMS state is unique (symmetric phase). This phase transition gives rise to baryon emergence (density-of-states divergence at the ζ pole).

2.3 Tensor Product Structure of the Dirac Operator

§2.1 established the angular structure: character χ_k labels the particle species, and the non-vanishing of $[D_F, \mu_g]$ produces gauge forces. §2.2 established the radial structure: $H = \log n$ labels energy and physical spatial position, and the pole of $\zeta(\beta)$ produces emergence and gravity.

A complete description of a particle requires both sets of information simultaneously: the angular structure determines the particle's identity (species, internal quantum numbers, including both direct labeling and color-singlet combinations), and the radial quantum number n determines the particle's energy scale (energy, spatial position). The two are independent—the angular part contains no energy scale (d_k is a dimensionless pure number), and the radial part does not distinguish particle species (H acts identically on all characters). The arithmetic Dirac operator D unifies both into a single structure.

Definition 2.2 (Arithmetic Dirac operator). The **arithmetic Dirac operator** is defined as

$$D = H \otimes I_F + \gamma \otimes D_F, \quad (19)$$

where H (§2.2) is the radial Hamiltonian (providing energy scale and KMS dynamics), D_F (§2.1) is the angular finite Dirac operator (whose character eigenspaces label particle species), I_F is the identity operator on the character space, and γ is the grading operator (satisfying $\gamma^2 = 1$, $\gamma D_F = -D_F \gamma$). Each fundamental particle state is jointly labeled by two sets of quantum numbers: the radial quantum number n (energy scale) and the angular quantum number k (identity). Matter states also include color-singlet combinations of fundamental states (baryons, expanded in §3.1).

The Dirac operator D is the complete dynamical operator of a particle. It simultaneously encodes the particle's energy evolution (radial H) and internal structure (angular D_F), unifying both into a single operator. The spectral data of D contain two layers of physical information: eigenvalues $(\log n + \gamma d_k)$ label fundamental particle states (fermions and gauge bosons), and the pole of the partition function $\zeta(\beta)$ labels collective emergent states (baryons). Together they constitute the complete spectral support of matter.

The tensor product structure $\otimes$ guarantees the independence of the two sets of quantum numbers: modifying the radial quantum number n (changing energy) does not alter the angular quantum number χ_k (particle identity is preserved); modifying the angular quantum number χ_k (changing particle species) does not alter the radial quantum number n (energy scale is preserved). This independence is a direct consequence of the tensor product form of D and requires no additional assumptions.

Proposition 2.3 (Spectral separation). *For positive integer n and character χ_k ,*

$$D |n, \chi_k\rangle = (\log n + \gamma d_k) |n, \chi_k\rangle. \quad (20)$$

Proof. $(H \otimes I_F + \gamma \otimes D_F) |n, \chi_k\rangle = (\log n) |n, \chi_k\rangle + \gamma d_k |n, \chi_k\rangle.$ □

Spectral separation has three direct physical corollaries:

- (i) **Identity conservation.** The angular quantum number χ_k is conserved under radial evolution $\sigma_t = e^{itH}$. H and the angular operator act on different factors of the tensor product, $[H \otimes I_F, I \otimes D_F] = 0$, so a particle's identity does not change under time evolution.
- (ii) **Equivalence principle.** The radial quantum number n does not distinguish particle species. $H|n\rangle = (\log n)|n\rangle$ is identical for all characters, so all particles respond identically in a gravitational field—gravity couples universally.
- (iii) **Angular eigenmodes have no intrinsic mass.** The angular eigenvalues d_k are dimensionless pure numbers and contain no energy scale; the radial H carries an energy scale but acts identically on all characters, without distinguishing particles. Therefore angular eigenmodes are intrinsically massless—mass can only be generated by a coupling term that simultaneously involves both tensor factors (§3.2).

Proposition 2.4 (Additive–multiplicative non-commutativity). $[D, \mu_g] = \gamma \otimes [D_F, \mu_g]$, whose matrix element in the χ_k direction is $d_k(\chi_k(g) - 1)$.

Proof. $[H \otimes I_F, \mu_g] = 0$ (radial commutes); hence $[D, \mu_g] = \gamma \otimes [D_F, \mu_g]$. When $\chi_0(g) \equiv 1$, $d_0(\chi_0(g) - 1) = 0$; when $\chi_k(g) \neq 1$ ($k \neq 0$), it is non-zero. $\square$

All non-commutativity comes from the angular D_F —the radial H commutes completely with μ_g . Physical meaning: gauge forces are generated entirely by the angular part; the radial part provides only the energy scale and spatial structure (gravity, emergence), and does not participate in gauge force generation. In the χ_0 direction the commutator vanishes, and the Dirac operator D reduces to purely radial: $D|n, \chi_0\rangle = (\log n)|n, \chi_0\rangle$. The physical significance of χ_0 (vacuum channel, Higgs mechanism) will be developed in §3.2.

2.4 KMS/Dirichlet Green Kernel

From the tensor product structure of the arithmetic Dirac operator $D = H \otimes I_F + \gamma \otimes D_F$, the square of the Dirac operator D is

$$D^2 = H^2 \otimes I_F + I \otimes D_F^2, \quad (21)$$

where the cross term $\{H \otimes I_F, \gamma \otimes D_F\} = 0$ (since H and D_F act on different factors of the tensor product, and $\gamma^2 = 1$). D^2 is diagonal in the eigenbasis $|n, \chi_k\rangle$:

$$D^2|n, \chi_k\rangle = [(\log n)^2 + d_k^2]|n, \chi_k\rangle, \quad (22)$$

where d_k is the k -th eigenvalue of D_F (§2.1). The character directions are completely decoupled, and the Green function $(D^2 - \lambda)^{-1}$ is block-diagonal in the character basis.

The Green kernels used in this paper take the heat-kernel form: the KMS trace $\text{Tr}_{\chi_k}(e^{-\beta H})$ on each character subspace serves as the propagator. By the decoupled structure of (22), these kernels naturally fall into two groups:

- χ_0 direction: D_F is frozen, driven purely by $H \rightarrow$ KMS partition function $\zeta(\beta)$;
- χ_k ($k \neq 0$) direction: D_F is active, twist insertion $\rightarrow$ Dirichlet L -function $L(\beta, \chi_k)$.

The vertex amplitude $|R_k|^2$ of gauge forces comes from the angular kernel, while the curvature constant C^* of gravity comes from the radial kernel.

Radial Green kernel. On the trivial-character χ_0 subspace, $D_F\chi_0 = d_0\chi_0$ and $[D_F, \mu_g]|_{\chi_0} = 0$; the angular degrees of freedom are frozen. The Dirac operator reduces to the purely radial Hamiltonian H , and the Green kernel simplifies to the KMS partition function [1]:

Proposition 2.5 (Radial KMS kernel). *The KMS trace (radial kernel) corresponding to the trivial character χ_0 is*

$$K_0(\beta) = \text{Tr}(e^{-\beta H}) = \sum_{n \geq 1} n^{-\beta} = \zeta(\beta). \quad (23)$$

$\zeta(\beta)$ has a unique simple pole at $\beta = 1$, corresponding to a divergence in the density of states, marking the KMS phase transition of the BC system from the disordered phase to the ordered phase. Gravity is therefore a purely radial force, originating from H rather than D_F .

Angular Green kernel. On the non-trivial character χ_k ($k = 1, \dots, 5$) subspaces, $[D_F, \mu_g]|_{\chi_k} \neq 0$, and the angular degrees of freedom are active. The twist operator U_{χ_k} inserts the angular information into the radial summation, yielding the Dirichlet L -function:

Proposition 2.6 (Angular Dirichlet kernel). *For non-trivial character χ_k ($k = 1, \dots, 5$),*

$$K_{\chi_k}(\beta) = \text{Tr}(U_{\chi_k} e^{-\beta H}) = \sum_{n \geq 1} \chi_k(n) n^{-\beta} = L(\beta, \chi_k). \quad (24)$$

$L(\beta, \chi_k)$ is regular and finite at $\beta = 1$ (since $\chi_k \neq \chi_0$). The L -function describes the propagation response in the character χ_k direction: a matter source produces a signal in the χ_k direction, which propagates via the L -function and is received by particles in the same character direction.

2.5 Vertex Amplitude R_k

The radial kernel diverges at $\beta = 1$ while the angular kernel is regular and finite—this contrast in analytic behavior is the mathematical origin of matter (§3), forces (§4), and coupling constants (§5) in subsequent sections.

The normalized response is defined as

$$R_{\chi_k}(\beta) = \frac{L(\beta, \chi_k)}{\zeta(\beta)}. \quad (25)$$

Physically, R_{χ_k} measures the response strength of a non-trivial channel relative to the vacuum. At the natural point $\beta = 1/2$ of the spectral action (corresponding to the critical line $s = 1/2$),

$$R_k \equiv R_{\chi_k}(1/2) = \frac{L(1/2, \chi_k)}{\zeta(1/2)}, \quad (26)$$

this gives the vertex Green amplitude for fermions and gauge couplings. The $|R_k|^2$ values for the five non-trivial characters are [13]

$$|R_1|^2 = |R_5|^2 = 0.344760, \quad |R_2|^2 = |R_4|^2 = 0.047562, \quad |R_3|^2 = 0.616448. \quad (27)$$

These numerical values enter directly into the subsequent mass formulas and coupling constant calculations.

Two-layer structure. R_k is not only the normalized response of the Green kernel, but also the vertex amplitude for the coupling of forces to matter. The BC system possesses a two-layer structure with no direct counterpart in continuous field theory: the classification of gauge fields (field sector, labeled by the adjoint eigenvalue λ_k of Δ_{mult}) and the coupling vertex of forces (matter vertex, labeled by the response kernel R_k of the matter character) originate from different mathematical objects.

The coupling vertex of each gauge force is given by the R_k of the matter characters participating in that force [13, 14]:

$$R_{\text{EM}} = R_2, \quad R_W = R_3, \quad R_s = R_1. \quad (28)$$

Interaction	Field sector (Δ_{mult} eigenvalue)	Matter vertex (R_k)	Source
Electromagnetic	$U(1)$ center/Cartan, $\lambda = 0$	R_2 (charge sector, $ R_2 ^2 = 0.048$)	XII + IV
Weak force	χ_3 root, $\lambda = 4$	R_3 (pure weak sector, $ R_3 ^2 = 0.616$)	XII + IV
Strong force	$\chi_2 \oplus \chi_4$ roots, $\lambda = 3$	R_1 (colored sector, $ R_1 ^2 = 0.345$)	XII + IV

The field sector is determined by the adjoint spectrum of Δ_{mult} (the multiplicative Laplacian)—answering which forces exist; the matter vertex is determined by R_k (the normalized response)—answering the coupling amplitude of forces. The two are independent, and ultimately converge in the CRT structure of $(\mathbb{Z}/7\mathbb{Z})^*$ [14]. Gravity and the nuclear force do not couple through character vertices; their structures will be developed separately in §4.3 (radial Green kernel) and §?? (radial–angular mixing). This two-layer structure will be used separately in §4 (force equations) and §5 (coupling constants).

3 Matter as Spectral Eigenmodes

§2 established the spectral structure of the arithmetic Dirac operator D (Definition 2.2). This section identifies matter as spectral eigenmodes of the Dirac operator D . In the series of works, the arithmetic description of matter has been established from two directions:

- [13] established the correspondence between fundamental fermions and angular characters: the CRT decomposition divides six characters into four physical sectors, and masses are uniformly given by L -function values on the critical line;
- [16] established the description of baryons as collective emergent states: the pole of $\zeta(\beta)$ at $\beta = 1$ causes the density of states to diverge in the color-singlet direction, and emergent states automatically possess color-singlet quantum numbers, integer charge, non-constituent mass, and absolute stability.

Definition 3.1 (Spectral eigenmode). A **fundamental excitation** in the spectrum of D is called a **spectral eigenmode** (particle). Each spectral eigenmode possesses:

1. **Angular position** (identity): its direction in character space, labeling what the particle is. χ_0 is the vacuum reference direction; baryons, as color singlets, project back to the χ_0 direction via ε^{abc} (quantum number inheritance).
2. **Radial position** (energy scale): its position in the radial spectrum, labeling how heavy the particle is. $N = 1$ is massless, $N > 1$ is massive, and the energy scale is given by $E = 1/\sqrt{G(N)}$.

Amplitude fluctuations of the condensate (Higgs) are collective oscillation modes, not fundamental excitations.

§2.3 proved that the commutator of D with μ_g differs essentially in different character directions (Proposition 2.4). This difference divides the physical roles of angular position and radial position into two classes:

3.1 Spectral Eigenmodes: Angular and Radial

Definition 3.2 (Spectral Eigenmode). A **fundamental excitation** in the spectrum of D is called a **spectral eigenmode** (particle). Each spectral eigenmode possesses:

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§2.3 proved that the commutator of D with μ_g has essential differences in different character directions (Proposition 2.4). This difference divides the physical roles of angular position and radial position into two classes:

Angular position $\rightarrow$ identity. Angular position determines particle identity. In non-trivial character (χ_k , $k \neq 0$) directions, $[D_F, \mu_g] \neq 0$, and $\lambda_k = d_k$ (§2.1 table) specifies the quantum numbers: the CRT decomposition $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ yields weak isospin (order 2), color (order 3), and weak $\times$ color (order 6). Three generations arise from three doublets D_1, D_2, D_3 of the involution. Angular eigenmodes are the fundamental fermions. The χ_0 direction provides color singlet identity: colored modes (χ_1/χ_5 directions) project to χ_0 via ε^{abc} , with quantum numbers given by the sum of constituents, i.e. baryons [16]. Angular position determines **what** the particle is.

Radial state number $N \rightarrow$ mass. N is the effective number of states of the radial Hamiltonian $H = \log n$, and the mass scale is given by $E = 1/\sqrt{G(N)}$. The fermion energy scale is determined by $N_{\text{EW}} = \varphi(p^4)$, propagating via the L -function path in each respective character direction [13]; the baryon energy scale is given by the divergence of the density of states at the pole of $\zeta(\beta)$ [16]. Emergence is strictly confined to χ_0 (the color singlet direction): in non-trivial character directions $L(1, \chi_k)$ is finite, and no emergence occurs—this is the arithmetic statement of color confinement. N determines **how heavy** the particle is.

The two classes of eigenmodes are summarized in the following table:

Eigenmode	Angular position	N	Spectral function	Matter
Angular	$\chi_{1-5} (d_k \neq 0)$	$\varphi(p^4)$, propagated via $L(\frac{1}{2}, \chi_k)$	$L(\beta, \chi_k)$	Fermions
Radial emergence	χ_0 (color singlet)	ζ pole emergence	$\zeta(\beta)$	Baryons

3.2 The Coupling Channel: Character χ_0

§2.3 proved that the tensor product structure of the Dirac operator strictly separates identity (angular) from energy scale (radial). §3.1 further classified matter on this basis into angular eigenmodes (fermions) and radial emergent states (baryons). A natural question arises: how do identity and energy scale, residing in two independent factors, coexist on the same particle?

The answer is the unique coupling channel: the trivial character χ_0 . χ_0 simultaneously satisfies two conditions—vanishing angular tension ($[D_F, T_a]|_{\chi_0} = 0$, carrying no quantum

numbers) and radial completeness ($L(\beta, \chi_0) = \zeta(\beta)(1-p^{-\beta})$, accessing all energy scales)—making it the unique information channel between the angular and radial sectors. The mass of a particle is the coupling strength of its state to the χ_0 condensate; no coupling means no mass.

The following subsections establish in turn the structure of χ_0 (§3.2.1), the inevitability of condensation (§3.2.2), the dynamics of condensation (§3.2.3), the uniqueness of the breaking direction (§3.2.4), and finally a complete characterization of matter (§3.3).

3.2.1 Properties of Character χ_0

This subsection proves the central theorem referenced at the beginning: χ_0 is the unique direction capable of carrying the angular–radial coupling.

Theorem 3.3 (Universality of χ_0). *The trivial character χ_0 is the unique direction in BC spectral dynamics that introduces no discrimination in either the angular or radial sector:*

1. **Angular universality.** $d_0 = 0$, $[D, \mu_g]|_{\chi_0} = 0$. *The Dirac operator does not distinguish any multiplicative group element in the χ_0 direction, reducing to purely radial:*

$$D |n, \chi_0\rangle = (\log n) |n, \chi_0\rangle. \quad (29)$$

2. **Radial universality.**

$$L(\beta, \chi_0) = \zeta(\beta) (1 - p^{-\beta}). \quad (30)$$

χ_0 does not distinguish any positive integer—its L -function sums over all n with equal weight, i.e. it is the partition function of the system itself.

3. **Normalization baseline.**

$$R_0(\beta) = \frac{L(\beta, \chi_0)}{\zeta(\beta)} = 1 - p^{-\beta} \xrightarrow{\beta \rightarrow \infty} 1. \quad (31)$$

The response kernel of χ_0 is the normalization reference for all other characters.

Proof. (1) By Proposition 2.3 with $k = 0$ and Proposition 2.4. (2) $\chi_0(n) = 1$ for all $(n, p) = 1$, hence $L(\beta, \chi_0) = \sum_{(n, p)=1} n^{-\beta} = \zeta(\beta)(1 - p^{-\beta})$. (3) From (2) divided by $\zeta(\beta)$. $\square$

The three statements of Theorem 3.3 correspond respectively to the threefold qualification of the coupling channel: angular universality $\Rightarrow$ carries no quantum numbers (can serve as background); radial universality $\Rightarrow$ accesses all energy scales (can supply energy); normalization baseline $\Rightarrow$ propagation in all other channels is referenced to χ_0 . Below we promote the first two statements to corollaries and add a remark on reality.

Corollary 3.4 (Gauge-singlet property). *For any automorphism σ of $(\mathbb{Z}/7\mathbb{Z})^*$, $\chi_0 \circ \sigma = \chi_0$; in particular, χ_0 is invariant under the CRT factor-preserving gauge group (§??) and carries no gauge charge.*

Proof. $\chi_0(g) \equiv 1$, and automorphisms merely permute group elements without changing a constant function. $\square$

Corollary 3.5 (Vanishing tension). *For any additive translation T_a ($a \in (\mathbb{Z}/7\mathbb{Z})^*$),*

$$[D_F, T_a]|_{\chi_0} = 0. \quad (32)$$

That is, addition and multiplication fully commute in the χ_0 direction, with no intrinsic tension.

Proof. $T_a\chi_0(g) = \chi_0(g+a) = 1 = \chi_0(g)$, hence $T_a\chi_0 = \chi_0$, and the two terms $D_F(T_a\chi_0)$ and $T_a(D_F\chi_0)$ are equal. $\square$

The non-trivial directions with $[D_F, T_a] \neq 0$ carry all the physics: gauge forces are angular deviations ($d_k \neq 0$), fermion masses are propagation along angular paths ($R_k \neq 1$), gravity is collective emergence at radial poles, and nuclear forces are joint angular–radial deviations. The twofold universality of χ_0 (vanishing angular tension + radial completeness) means it carries no quantum numbers yet accesses all energy scales—precisely the defining characteristic of the physical vacuum. The condensate of χ_0 is the vacuum itself.

3.2.2 Inevitability of Condensation

$\zeta(\beta)$ has a pole at $\beta = 1$, the density of states diverges, the symmetric vacuum is unstable, and condensation is a natural outcome.

Proposition 3.6 (Inevitability of condensation). *The VEV in the χ_0 direction is non-zero, as established by two independent paths:*

1. **Potential instability.** Define the free energy in the χ_0 direction as $F_0(\beta) = -\log \zeta(\beta)$. From the Laurent expansion $\zeta(\beta) = (\beta - 1)^{-1} + \gamma + O(\beta - 1)$,

$$F_0''(\beta) = -\frac{\zeta''(\beta)}{\zeta(\beta)} + \left(\frac{\zeta'(\beta)}{\zeta(\beta)}\right)^2 \xrightarrow{\beta \rightarrow 1^+} -\frac{1}{(\beta - 1)^2} < 0. \quad (33)$$

The effective potential has negative curvature at the symmetric point, corresponding to an unstable maximum; the system necessarily evolves to a condensed state;

2. **Occupation dominance.** The normalized response of the KMS state on characters is $R_k(\beta) = L(\beta, \chi_k)/\zeta(\beta)$ [12]. At the critical line $\beta = 1/2$, the Gauss sum gives

$$R_0 = 1 - \frac{1}{\sqrt{p}} = 1 - \frac{1}{\sqrt{7}} = 0.622 > \frac{1}{2}, \quad (34)$$

i.e. the occupation weight of the χ_0 direction exceeds one half ($R_0 > 1/2 \iff p > 4$, which holds for $p = 7$).

Proof. (1) Substituting $\zeta(\beta) = (\beta - 1)^{-1} + \gamma + O(\beta - 1)$ into the expression for F_0'' , the leading term $-1/(\beta - 1)^2$ is negative and divergent as $\beta \rightarrow 1^+$. (2) By the Euler product, $L(\beta, \chi_0) = \zeta(\beta)(1 - p^{-\beta})$ (Theorem 3.3(2)); at $\beta = 1/2$, $|R_0|^2 = (1 - p^{-1/2}) \cdot (\text{normalization factor})$, substituting $p = 7$ yields 0.622. $\square$

Remark 3.7. The two paths are logically independent: (1) is thermodynamic (free energy curvature), (2) is spectral (occupation probability); both point to $\text{VEV} \neq 0$. The inevitability of condensation is thereby established; the manner in which it occurs is determined by the KMS structure, as shown in the next subsection.

3.2.3 Dynamics of Condensation: Phase Transition at $\beta = 1$

Theorem 3.8 (KMS phase transition [1]). *The KMS state structure of the BC system undergoes a phase transition at $\beta = 1$:*

1. $0 < \beta \leq 1$: the KMS state is unique, and $\langle \chi_k \rangle_\beta = 0$ for all $k \neq 0$ (symmetric phase);
2. $\beta > 1$: the extremal KMS states form a continuous family $\{\varphi_{\beta, \rho} : \rho \in \hat{\mathbb{Z}}^*\}$, and $\langle \chi_k \rangle_{\beta, \rho} \neq 0$ (spontaneous symmetry breaking).

The phase transition point coincides with the pole of $\zeta(\beta)$.

§3.2 established that mass originates from coupling to χ_0 ; Theorem 3.8 is its dynamical realization: the VEV transitions from zero to non-zero, and the condensate forms. The two sides of the same phase transition produce two classes of spectral support (echoing the matter classification of §3.1):

1. **Cold side** (frozen phase, $\beta \gg 1$): the χ_0 condensate carries $N_{\text{EW}} = \varphi(7^4) = 2058$ vacuum modes, giving the electroweak scale v_{EW} via (4); after breaking along χ_3 (§3.2.4), it gives masses to W/Z and fermions;
2. **Critical side** ($\beta \rightarrow 1^+$): the ζ pole causes the density of states to diverge, color singlet collective emergence occurs, and the state number $N_{\text{baryon}} = \zeta(\beta) \approx 2.35$ gives the baryon scale 937 MeV.

The cold side and critical side share the same scale formula (4), differing only in the value of the state number N : the cold side uses algebraic counting (integer $\varphi(p^4)$), while the critical side uses the partition function ($\zeta(\beta)$).

After the condensate forms, amplitude fluctuations of the order parameter in the χ_0 direction exist around the new ground state. This fluctuation mode is the experimentally observed Higgs particle. However, it is not a particle in the sense of spectral eigenmodes (Definition 3.2). Although the Higgs resides in the χ_0 direction (has an angular position), it does not satisfy the **independence** condition: it is not a fundamental excitation in the spectrum of D , but rather the amplitude fluctuation of the χ_0 condensed state around the new ground state—a collective oscillation mode. By contrast: baryons also reside in the χ_0 direction, but they are fundamental excitations emerging at the ζ pole; the Higgs is not a new excitation, only a fluctuation of the pre-existing condensed state.

Observable symmetry breaking requires projection along a non-trivial direction—the next subsection proves that this direction is unique.

3.2.4 Uniqueness of the Breaking Direction: χ_3

Lemma 3.9 (Unique non-trivial real character). *In the character group of $(\mathbb{Z}/7\mathbb{Z})^*$, χ_3 is the unique non-trivial self-conjugate (real) character.*

Proof. $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}$, and its character group $\widehat{G} \cong \mathbb{Z}/6\mathbb{Z}$ consists of $\chi_0, \dots, \chi_5$. The self-conjugacy condition $\chi_k = \bar{\chi}_k$ is equivalent to $\chi_k = \chi_{6-k}$, i.e. $2k \equiv 0 \pmod{6}$, with solutions $k \in \{0, 3\}$. χ_0 is the trivial character, so χ_3 is the unique non-trivial real character. Equivalently, χ_3 is the unique element of order 2 in $\widehat{G}$, corresponding to the Legendre symbol $(\frac{\cdot}{7})$. $\square$

Theorem 3.10 (Unique condensation direction). *On the KMS states of BC , the unique real order parameter direction that can break gauge symmetry is χ_3 .*

Proof. Let φ_β be the KMS state at temperature β .

1. φ_β is a positive functional satisfying $\varphi_\beta(x^*) = \overline{\varphi_\beta(x)}$.
2. The adjoint of a character is its complex conjugate: $\chi_k^* = \bar{\chi}_k = \chi_{6-k}$.
3. The order parameter $\langle \chi_k \rangle = \varphi_\beta(\chi_k)$ is real if and only if χ_k is self-conjugate.
4. By Lemma 3.9, the only non-trivial self-conjugate character is χ_3 .
5. χ_0 is trivial ($\chi_0(g) \equiv 1$); its condensation does not break any gauge symmetry.

Hence χ_3 is the unique direction that can produce a real order parameter and thereby break gauge symmetry. $\square$

Remark 3.11 (Dynamical consistency). The algebraic uniqueness and dynamical analysis point to the same direction: $|L(1, \chi_3)|$ is the largest among all non-trivial characters, so the χ_3 direction condenses first as $\beta \rightarrow 1^+$.

Corollary 3.12 (Masses of three types of gauge bosons). *By Theorem 3.10 and Lemma 3.9:*

- **Weak** $SU(2)$ (χ_3 direction, real, largest $|L(1)|$): the unique breaking direction; condensation produces a real order parameter $\Rightarrow W/Z$ are massive. The reality of χ_3 guarantees $\arg L(1/2, \chi_3) = 0$ (mass is real and positive); the reality factor $c_3 = 1 + [\chi_3 = \bar{\chi}_3] = 2$ doubles the weak channel, giving $N_3 = |\ker \chi_3| \cdot d_2 \cdot c_3 = 18$.
- **Color** $SU(3)$ ($\chi_2 \oplus \chi_4$, complex conjugate pair): no real order parameter $\Rightarrow$ no breaking $\Rightarrow$ gluons are massless.
- **Electromagnetic** $U(1)$ (χ_0 , trivial): the order parameter is gauge-invariant $\Rightarrow$ no breaking $\Rightarrow$ the photon is massless.

The uniqueness of the breaking direction divides gauge bosons into three classes, summarized in the following table:

Direction	Character	Coupling to χ_0	Broken?	Boson mass
Weak $SU(2)$	χ_3 (real)	Yes	Broken	W/Z massive
Color $SU(3)$	$\chi_2 \oplus \chi_4$ (complex)	No	Unbroken	Gluons massless
Electromagnetic $U(1)$	χ_0 (trivial)	—	Unbroken	Photon massless

3.3 Spectral Eigenmode Summary Table

The KMS condensed state of χ_0 defines the physical vacuum. The two fundamental attributes of matter—identity and mass—are both determined relative to this vacuum background:

- **Identity** (angular position): the χ_0 condensed state provides the reference direction; fundamental fermions are determined by angular displacements relative to χ_0 (χ_1 – χ_5); baryons, as color singlets, project back to the χ_0 direction via ε^{abc} ;
- **Mass** (radial): fermions acquire mass through propagation along radial paths coupled to the χ_0 condensate, with scale given by $G(N)$; baryon mass is the radial scale of the χ_0 direction itself (m_p); no coupling to χ_0 means no mass.

The Dirac operator $D = H \otimes I_F + \gamma \otimes D_F$ unifies these two aspects: angular position gives identity, radial position gives energy scale. Matter particles are fundamental excitations in the spectrum of D —each Standard Model particle corresponds to a definite (angular position, radial position) pair. The following table lists both for each particle:

Particle	Identity	Energy scale	Spectral function	Mechanism
γ	χ_0 Cartan direction	No coupling, $N = 1$	—	Gauge invariance
g	$\chi_2 \oplus \chi_4$ (color charge)	No coupling, $N = 1$	—	Color confinement
W/Z	χ_3 (weak isospin)	$N = \varphi(p^4)$	$L(\beta, \chi_3)$	χ_3 condensation breaking
12 fermions	χ_{1-5} CRT coordinates	$N = \varphi(p^4)$, propagated via $L(\frac{1}{2}, \chi_k)$	$L(\beta, \chi_k)$	Angular path propagation
Baryons	Color singlet combination	ζ pole emergence	$\zeta(\beta)$	Collective emergence

The Higgs particle is not included in the above table: although it resides in the χ_0 direction (has an angular position), it is not a fundamental excitation but a collective oscillation mode (§3.2.3).

4 Interactions as Green Responses

The Standard Model contains three gauge forces, gravity, and the nuclear force. Although these forces correspond to different character directions and operators, they share a unified mathematical description in BC: every force is a response of the Green function $(D^2 - \lambda)^{-1}$ of D in the corresponding direction.

4.1 Angular/Radial Decomposition of the Green Kernel

From the tensor product structure of the arithmetic Dirac operator $D = H \otimes I_F + \gamma \otimes D_F$, the Green function $(D^2 - \lambda)^{-1}$ of the Dirac operator naturally separates into two families of kernels in the character basis.

In the vacuum direction (χ_0) , $[D_F, \mu_g] = 0$ (Proposition 2.4), the Dirac operator reduces to pure H , and the angular part D_F plays no role. The corresponding Green kernel is the KMS partition function

$$K_0(\beta) = \text{Tr } e^{-\beta H} = \zeta(\beta). \quad (35)$$

$\zeta(\beta)$ controls the collective response in the vacuum direction: its pole at $\beta = 1$ produces a divergence in the density of states (baryon emergence), and the radial potential $H = \log n$ satisfies a discrete Poisson equation (gravity). Gravity is therefore a purely radial force, arising from H rather than D_F .

In the angular directions (nontrivial characters χ_k , $k \neq 0$), $[D_F, \mu_g] \neq 0$, and D_F is nontrivial. The corresponding Green kernels are Dirichlet L -functions

$$K_{\chi_k}(\beta) = \text{Tr}(U_{\chi_k} e^{-\beta H}) = L(\beta, \chi_k). \quad (36)$$

$L(\beta, \chi_k)$ describes the propagation response in the character χ_k direction: a matter source generates a signal in the χ_k direction, which propagates via the L -function and is received by particles in the same character direction. $\zeta(\beta)$ as the denominator provides normalization; the physical response kernel is $R_k = L(\beta, \chi_k)/\zeta(\beta)$. The three gauge forces—electromagnetic (χ_0 center, $\lambda = 0$), weak (χ_3 root, $\lambda = 4$), and color ($\chi_2 \oplus \chi_4$ root, $\lambda = 3$)—are all propagated by the angular Green kernel $L(\beta, \chi_k)$.

Furthermore, the nuclear force is a joint angular–radial phenomenon: the transition between two baryons transmits a π meson via an angular vertex in the χ_1 direction, i.e. emergent state $\rightarrow \chi_1$ (angular transition) $\rightarrow \chi_0$ (baryon). The nuclear force therefore involves both families of Green kernels simultaneously, serving as a direct manifestation of the angular $\times$ radial combination.

The above analysis is summarized in the following table.

Force	Direction	Green kernel	Physical mechanism
EM/weak/strong	Angular	$L(\beta, \chi_k)$	Propagation response in character direction
Gravity	Radial	$\zeta(\beta)$	Collective response of $H = \log n$
Nuclear	Angular $\times$ radial	$L \otimes \zeta$	$\chi_0 \rightarrow \chi_1 \rightarrow \chi_0$ transition

The three classes of forces correspond respectively to the three modes of the Dirac operator’s tensor product structure: purely angular, purely radial, and angular–radial hybrid.

4.2 Discrete Gauge Equations

The continuous electromagnetic field is completely characterized by Maxwell's equations: two constraint equations ($\nabla \cdot \mathbf{B} = 0$ and $\nabla \cdot \mathbf{E} = \rho$), two dynamical equations (Faraday and Ampère), and charge conservation as a corollary. The internal space of the BC system is the discrete $C_6 = (\mathbb{Z}/7\mathbb{Z})^*$, yet it likewise supports a complete set of gauge equations. The key is that the BC system possesses *two types* of intrinsic derivatives, which respectively play the roles of spatial and temporal derivatives in Maxwell's theory:

- **Angular derivative** $D_a = S_a - I$, where S_a is the action of the multiplicative shift μ_g along CRT coordinate direction a —this gives the internal gauge structure;
- **KMS flow** $\sigma_t = e^{itH}$, the modular evolution generated by $H = \log n$ —this gives the temporal dynamics.

Maxwell's single spacetime derivative ∂_μ splits in the BC system into two types, angular (internal) and KMS (temporal), reflecting the refinement of discrete arithmetic structure relative to continuous field theory. The equations are given below in five groups, with each corresponding to its Maxwell counterpart.

(I) Sector eigenvalue equations. Let $L_{\text{ang}} := -\Delta_{\text{mult}}$ be the angular multiplicative Laplacian acting on C_6 , and let $P_Y, P_W, P_c, P_{\text{mix}}$ be the spectral projections onto the $U(1)$ center, weak root, color root, and weak–color mixed sectors respectively. The allowed gauge fields are classified by eigenvalue:

$$L_{\text{ang}} A_Y = 0, \tag{37}$$

$$L_{\text{ang}} A_W = 4A_W, \tag{38}$$

$$L_{\text{ang}} A_c = 3A_c. \tag{39}$$

The electromagnetic/hypercharge field is the center–Cartan zero mode ($\lambda = 0$); the weak field is a root mode that changes only the $\mathbb{C}^2$ factor ($\lambda = 4$); the color field is a root mode that changes only the $\mathbb{C}^3$ factor ($\lambda = 3$).

(II) Constraint equation (corresponding to $\nabla \cdot \mathbf{B} = 0$). Define the discrete field strength

$$F = D_2 A_3 - D_3 A_2, \tag{40}$$

where D_2, D_3 are the angular differences in the weak ($\mathbb{C}^2$) and color ($\mathbb{C}^3$) directions respectively. Since $C_6 \cong C_2 \times C_3$ is a CRT direct product, the multiplicative shifts in the two directions commute, $[D_2, D_3] = 0$ (verified numerically to be exactly zero), so the Bianchi identity

$$D_2 F_3 + D_3 F_2 = 0 \tag{41}$$

holds automatically. The constraint specific to the BC system is the CRT factor-preserving condition

$$P_{\text{mix}} F = 0, \tag{42}$$

meaning the projection of the field strength onto the weak–color mixed direction ($\lambda = 1$, $\text{dlog } k = 1, 5$) must vanish. This is the arithmetic counterpart of $\nabla \cdot \mathbf{B} = 0$: just as magnetic monopoles are excluded topologically, weak–color mixed gauge fields are excluded by the factor-preserving automorphism principle.

(III) Source equation: discrete wave equation. Define the angular potential

$$V_\chi(\beta) := \sum_{n \geq 2} \Lambda(n) [\chi(n) - 1] n^{-\beta}, \quad (43)$$

where $\Lambda(n)$ is the von Mangoldt function. The propagation of gauge field A_X at energy scale β is described by the sourced wave equation:

$$\boxed{\frac{\partial A_X}{\partial \beta} + V_{\chi X}(\beta) A_X = J_X(\beta), \quad X = Y, W, c.} \quad (44)$$

The Green function solution is

$$A_X(\beta) = R_\chi(\beta) \int_{\beta_0}^{\beta} \frac{J_X(\beta')}{R_\chi(\beta')} d\beta', \quad R_\chi = \frac{L(\beta, \chi)}{\zeta(\beta)}. \quad (45)$$

Physical consequences of the angular potential V_χ :

- $V_{\chi_0} \equiv 0$: the electromagnetic field has no angular potential, is massless, and propagates at long range.
- $V_{\chi_3} \neq 0$: the weak force field acquires an effective mass and is short-range.
- As $\beta \rightarrow 1^+$, $V_{\chi_2} \rightarrow \infty$: the color field potential diverges, corresponding to confinement.

Decreasing β corresponds to increasing energy scale; the solution propagates downward from β_{source} , forming a retarded kernel.

(IV) KMS dynamics. The modular time evolution of the gauge field is given by the Hamiltonian:

$$\frac{dA}{dt} = i[H, A], \quad [H, A]_{mn} = \log(m/n) A_{mn}, \quad (46)$$

The KMS frequency $\omega_{mn} = \log(m/n)$ is determined by the prime energy level differences. The homogeneous version of wave equation (44) is the evolution equation for R_χ :

$$\frac{d}{d\beta} \log R_\chi(\beta) = - \sum_{n \geq 2} \frac{\Lambda(n) [\chi(n) - 1]}{n^\beta}. \quad (47)$$

That is, the rate of change of the gauge response with KMS temperature equals the character-weighted prime power spectrum.

(V) Conservation law and gauge invariance. The character selection rule $\chi_{\text{in}} = \chi_{\text{out}}$ at each vertex gives angular momentum conservation, corresponding to charge conservation (continuity equation); its underlying basis is the prime flow conservation of the Euler product $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$. The BC counterpart of gauge invariance $A \rightarrow A + \nabla \lambda$ is the zero mode in the χ_0 direction: by $\lambda_0 = 0$ (37), translations along χ_0 do not change the field strength.

Equations (37)–(42) determine the allowed gauge field sectors, (44) determines the Green response to matter sources, and (47) gives the KMS temporal dynamics. The discrete gauge equations are summarized in the following table.

Equation	Content	Physical meaning
(37)–(39)	$L_{\text{ang}} A_X = \lambda_X A_X$	Gauge fields classified by eigenvalue
(42)	$P_{\text{mix}} F = 0$	Weak–color mixing forbidden
(44)	$\partial_\beta A_X + V_\chi A_X = J_X$	Discrete wave equation
(47)	$\frac{d}{d\beta} \log R_\chi = - \sum \Lambda(n) [\chi(n) - 1] n^{-\beta}$	Green kernel evolution
(46)	$\omega_{mn} = \log(m/n)$	Discrete gauge wave frequency
—	$\chi_{\text{in}} = \chi_{\text{out}}$	Internal quantum number conservation

Remark 4.1 (Physical space representation of gauge forces). The gauge force equation (44) describes force propagation in the physical space $\mathbb{N}^*$. β is the Mellin dual variable (momentum representation) of $\mathbb{N}^*$, and the Green kernel $R_\chi(\beta) = L(\beta, \chi)/\zeta(\beta)$ is the propagator of the gauge force.

4.2.1 Non-Abelian Self-Coupling

Equation (??) is independent for each sector, corresponding to Abelian gauge fields. The non-Abelian self-coupling of the weak and strong forces is given by character multiplication:

$$\chi_i \cdot \chi_j = \chi_{i+j \bmod 6}.$$

Define the non-Abelian field strength

$$F_k = D_a A_k + \sqrt{\alpha_k} \sum_{i+j \equiv k} f_{ijk} A_i A_j, \quad (48)$$

with structure constants

$$f_{ijk} = \delta_{i+j, k \bmod 6} \cdot \varepsilon(i, j), \quad \varepsilon(i, j) = -\varepsilon(j, i), \quad (49)$$

uniquely determined by character multiplication [14]. The character group $(\widehat{\mathbb{Z}/7\mathbb{Z}})^* \cong \mathbb{Z}/6\mathbb{Z}$ has Abelian multiplication, but after the CRT factor-preserving constraint projects it onto the $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ sectors, the self-coupling structure within each sector reproduces the non-Abelian nature of $SU(2)$ and $SU(3)$ respectively: $\chi_3 \cdot \chi_3 = \chi_0$ gives the $SU(2)$ pattern $2 \times 2 \rightarrow 1$, and $\chi_2 \cdot \chi_2 = \chi_4$ gives the $SU(3)$ cyclic pattern $3 \times 3 \rightarrow \bar{3}$. The coupling strengths $\sqrt{\alpha_k}$ are given by the Born rule of §5. Verification sector by sector:

- χ_0 direction: $\chi_0 \cdot \chi_j = \chi_j$, no new direction is produced, $f_{00k} = 0$ — no self-coupling (Abelian). ✓
- χ_3 direction: $\chi_3 \cdot \chi_3 = \chi_0$, two weak force modes couple to produce the neutral direction. ✓
- χ_2 direction: $\chi_2 \cdot \chi_2 = \chi_4 = \bar{\chi}_2$, three color directions couple cyclically $\chi_2 \rightarrow \chi_4 \rightarrow \chi_0 \rightarrow \chi_2$. ✓

Substituting (48) into (??) yields the complete non-Abelian gauge equation:

$$\boxed{\frac{\partial A_k}{\partial \beta} + V_{\chi_k}(\beta) A_k + \sqrt{\alpha_k} \sum_{i+j \equiv k} f_{ijk} A_i A_j = J_k(\beta).} \quad (50)$$

The linear part describes propagation and mass; the nonlinear part describes self-coupling; both are unified within the same β -flow equation.

4.3 Gravity, Baryons, and Nuclear Force

Gravity, baryon mass, and the nuclear force—three seemingly unrelated physical quantities—all arise from the same radial structure of the BC system: the positive integer energy spectrum controlled by the Hamiltonian $H = \log n$.

4.3.1 Gravity Equations

Gravity is a purely radial force, controlled by the ζ -function structure of the Hamiltonian $H = \log n$. In parallel with the purely angular equation system for gauge forces (§4.2), gravity likewise possesses a complete equation structure.

(I) Sector eigenvalue equation: zero mode. Gravity corresponds to the $\lambda_0 = 0$ zero mode of the angular Laplacian Δ_{mult} —the χ_0 direction.

$$L_{\text{ang}} \Phi = 0. \quad (51)$$

The gravitational potential Φ is completely flat in the angular directions (zero eigenvalue); all angular structure is controlled solely by the radial Hamiltonian $H = \log n$. This contrasts with gauge forces having nonzero eigenvalues ($\lambda = 3, 4$): gauge forces “oscillate” in the angular direction, while gravity “rests” in the angular direction.

(II) Constraint equation: equivalence principle. $\chi_0(g) = 1$ for all g —the vacuum direction treats all characters equally. Therefore gravity couples identically to all particles, regardless of color, flavor, or weak isospin:

$$\langle \chi_k | \Phi | \chi_k \rangle = \langle \chi_j | \Phi | \chi_j \rangle \quad \forall j, k. \quad (52)$$

This is the arithmetic statement of the equivalence principle (universality of gravitational coupling). For comparison: the gauge force constraint $P_{\text{mix}} F = 0$ *forbids* certain sectors; the gravity constraint *requires* all sectors to be equivalent. Both constraints originate from the special nature of χ_0 —gauge forces through exclusion, gravity through universality.

(III) Field equation: discrete Poisson. In continuous spacetime, gravity is described by the Poisson equation $\nabla^2 \Phi = 4\pi G \rho$. The radial potential $H(n) = \log n$ of the BC system satisfies the discrete Poisson equation [15]

$$-\Delta(\log n) = R(n) = \log \frac{n^2}{n^2 - 1} = n^{-2} + O(n^{-4}), \quad (53)$$

where Δ is the radial projection of the multiplicative Laplacian on $\mathbb{N}^*$. The discrete curvature $R(n) \sim n^{-2}$ is the arithmetic version of the gravitational $1/r^2$ force, with the positive integer label n playing the role of discrete distance. The discrete curvature constant

$$C^* = 2 \sum_{n \geq 1} \log^2(1 + 1/n) = 1.954378 \quad (54)$$

is the sum of curvature squares over all sites on $\mathbb{N}^*$, and it enters both the gravitational coupling (§??) and the baryon mass (§4.3).

(IV) KMS dynamics: temperature evolution of the ζ -function. The evolution of the gravitational response kernel $K_0(\beta) = \zeta(\beta)$ with inverse temperature β is

$$\frac{d}{d\beta} \log \zeta(\beta) = - \sum_{n \geq 2} \frac{\Lambda(n)}{n^\beta}, \quad (55)$$

where $\Lambda(n)$ is the von Mangoldt function. This has the same form as gauge force equation (47), but with the character taken as χ_0 (trivial), so the $\chi_0(n) - 1 = 0$ terms vanish; the right-hand side is a prime power sum *without character weighting*—gravity “feels” all primes without distinguishing any direction. As $\beta \rightarrow 1^+$, $\zeta'(\beta)/\zeta(\beta) \rightarrow -1/(\beta - 1)$, and the pole drives collective emergence.

(V) Conservation law: energy conservation. The gravitational field equation (53) maintains energy conservation:

$$\sum_{n \geq 1} R(n) = C^*/2 = \text{finite constant}. \quad (56)$$

The total curvature summed over all of space is finite and does not diverge—the discrete $\mathbb{N}^*$ naturally provides an infrared cutoff. This corresponds to the character conservation $\chi_{\text{in}} = \chi_{\text{out}}$ in gauge forces: gauge forces preserve angular quantum numbers, while gravity preserves total radial energy.

Summary of gravity equations.

Equation	Content	Physical meaning
(51)	$L_{\text{ang}}\Phi = 0$	Gravity is the angular zero mode, treating all particles equally
(52)	$\langle \chi_k \Phi \chi_j \rangle$ independent of k	Equivalence principle
(53)	$-\Delta(\log n) = R(n) \sim n^{-2}$	Discrete Poisson equation
(55)	$\frac{d}{d\beta} \log \zeta = - \sum_n \Lambda(n) n^{-\beta}$	KMS temperature evolution of gravitational response
(54)	$C^* = 2 \sum_n \log^2(1 + 1/n)$	Discrete curvature constant

Remark 4.2 (Physical space representation of gravity). The discrete Poisson equation (53) is defined directly on the physical space $\mathbb{N}^*$. n is the radial distance coordinate (additive metric $d_+(m, n) = |m - n|$), and $R(n) \sim 1/n^2$.

4.3.2 Nuclear Force Equations

The nuclear force is a joint angular–radial phenomenon: the initial and final states are baryons, and the transition vertex is the Dirichlet response of character χ_1 (angular character). Unlike the purely angular equation system for gauge forces (§4.2) and the purely radial equations for gravity (§4.3.1), Unlike the purely angular equations of gauge forces (§4.2) and the purely radial equation of gravity (§4.3.1), the nuclear force equations naturally embody the joint angular×radial structure. The following five sets of equations are structurally symmetric with the gauge force equations, but each one reflects the angular–radial coupling.

(I) Sector eigenequation. The nuclear force field simultaneously carries angular (χ_1) and baryon quantum numbers. The angular component satisfies

$$L_{\text{ang}} \Pi = \lambda_1 \Pi = 1 \cdot \Pi, \quad (57)$$

i.e. the pion field Π is the $\lambda = 1$ eigenmode of the angular Laplacian Δ_{mult} (CRT coordinate difference $(1, 1)$, corresponding to $\chi_1 \oplus \chi_5$). Note that $\lambda = 1$ is forbidden in gauge forces by the factor-preservation principle (§2.1.1), but the nuclear force is not a gauge force—it is an *effective* interaction between emergent states, not subject to the CRT factor-preservation constraint. The radial component is labeled by the quantum number n (color-singlet level) of the baryon state:

$$\Pi_{n_i \rightarrow n_f} : \quad n_i, n_f \geq 3, \quad (58)$$

transitions occur only between emergent states with $n \geq 3$ ($n = 1, 2$ have no emergence, no baryons).

(II) Constraint equation: color-singlet selection rule. The initial and final states of the nuclear force *must* be color singlets (χ_0 direction):

$$P_{\chi_0} |B_i\rangle = |B_i\rangle, \quad P_{\chi_0} |B_f\rangle = |B_f\rangle. \quad (59)$$

Colored states ($\chi_j, j \neq 0$) do not emerge (Corollary ??), and therefore do not participate in the nuclear force. This corresponds to the $P_{\text{mix}} F = 0$ constraint in gauge forces: gauge forces forbid weak–color mixed sectors, the nuclear force forbids non-color-singlet participation. Both constraints originate from the special nature of the χ_0 direction, but act at different levels—gauge forces constrain the adjoint sector of the field, the nuclear force constrains the emergent states of the source.

(III) Source equation: joint angular–radial transition. The transition amplitude between two baryon states $|B_i\rangle$ (χ_0 emergent) is

$$\mathcal{M}(B_i \rightarrow B_f + \pi) = \langle B_f | R_{\chi_1}(\beta) | B_i \rangle, \quad (60)$$

where $R_{\chi_1} = L(\beta, \chi_1)/\zeta(\beta)$ is the normalized response kernel of the χ_1 sector. The baryon properties of the initial and final states couple with the χ_1 property (angular) of the transition vertex within the same amplitude. This is a structure unique to the nuclear force: in the source equation (44) of gauge forces, both the field sector and the matter vertex are in the angular direction; the source of the nuclear force spans two directions.

(IV) KMS dynamics: temperature evolution of the transition kernel. The evolution of the transition response kernel $R_{\chi_1}(\beta)$ with inverse temperature β is given by the same KMS equation as for gauge forces:

$$\frac{d}{d\beta} \log R_{\chi_1}(\beta) = - \sum_{n \geq 2} \frac{\Lambda(n) [\chi_1(n) - 1]}{n^\beta}, \quad (61)$$

where $\Lambda(n)$ is the von Mangoldt function. This has the same form as gauge force equation (47), but the character changes from χ_3 (weak) or χ_2 (color) to χ_1 (quark sector). The order of χ_1 is 6 (fully covering $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$), making the prime summation on the right-hand side simultaneously involve both the weak and color directions—again manifesting the angular–radial coupling nature of the nuclear force.

(V) Conservation law: baryon number conservation. Nuclear force transitions preserve baryon number:

$$\Delta N_B = 0 : \quad B_i \rightarrow B_f + \pi \quad (N_B = 1 \rightarrow 1 + 0). \quad (62)$$

The pion is a $\chi_1 \bar{\chi}_1 = \chi_0$ composite state (baryon number zero), so baryon number is conserved across the transition. This corresponds to character conservation $\chi_{\text{in}} = \chi_{\text{out}}$ in gauge forces: gauge forces preserve character quantum numbers, the nuclear force preserves baryon number. The underlying reason is the same—prime flow conservation in the Euler product $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$.

(VI) Internal propagation: inclusive summation of the affine group. The internal propagation of the transition proceeds via the inclusive trace of $\text{Aff}(\mathbb{F}_7)$:

$$\eta_N = |\text{Aff}(\mathbb{F}_7)|_{\text{eff}} = \frac{42}{\tau_L}, \quad (63)$$

where $\tau_L = 3/(2|\zeta(1/2)|) = 1.027$ is the Loewy chain factor (§??). $\text{Aff}(\mathbb{F}_7)$ is the affine group over $\mathbb{F}_7$, with $|\text{Aff}(\mathbb{F}_7)| = 7 \times 6 = 42$; the τ_L correction arises from the non-semisimple (Loewy) structure of the group algebra. In contrast to gauge forces where $\eta = 1/N$ (path averaging, dilution), the nuclear force η *sums* over internal affine degrees of freedom (inclusive summation, enhancement). This contrast—averaging vs. summation—is the structural root cause of why gauge forces are weak ($\alpha \sim 10^{-2}$) while the nuclear force is strong ($\alpha_N \sim 14$). The numerical values of the nuclear coupling and pion mass are given in §??.

Summary of nuclear force equations.

Equation	Content	Physical meaning
(57)	$L_{\text{ang}} \Pi = \Pi \ (\lambda = 1)$	Nuclear force field is angular χ_1 eigenmode
(59)	$P_{\chi_0} B\rangle = B\rangle$	Initial/final states must be color singlets (baryons)
(60)	$\mathcal{M} = \langle B_f R_{\chi_1} B_i \rangle$	Joint angular–radial transition (source equation)
(61)	$\frac{d}{d\beta} \log R_{\chi_1}$	KMS temperature evolution (same form as gauge forces)
—	$\Delta N_B = 0$	Baryon number conservation
(63)	$\eta = 42/\tau_L$	Inclusive summation over internal affine degrees of freedom

Comparison of three types of forces. Gauge forces, gravity, and the nuclear force are all derived from Green’s functions of the same arithmetic Dirac operator D (Definition 2.2), but their propagation mechanisms differ fundamentally.

	Direction	Green kernel	Propagation mechanism	Constraint
Gauge forces	Angular	$L(\beta, \chi_k)$	Particle exchange ($[D_F, \mu_g] \neq 0$ produces propagator)	$P_{\text{mix}} F = 0$
Gravity	Radial	$\zeta(\beta)$	Collective potential ($[D_F, \mu_g] = 0$, no propagator)	Equivalence principle
Nuclear force	Angular×Radial	$L \otimes \zeta$	Coupled transition ($\chi_0 \rightarrow \chi_1 \rightarrow \chi_0$)	Color singlet

The core distinction among the three types of forces lies in their propagation mechanisms. Gauge forces produce propagators (gauge bosons) through $[D_F, \mu_g] \neq 0$ in the angular character direction; particles interact by exchanging propagators. Gravity has no propagation medium: in the χ_0 direction $[D_F, \mu_g] = 0$, there is no angular character that can serve as a propagator; gravity acts directly on all particles through the discrete Poisson

equation of the radial potential $H = \log n$ — this is precisely the arithmetic origin of the equivalence principle (gravity couples universally to all particles). The nuclear force lies between the two: the initial and final states are baryons, but the transition exchanges pions via an angular vertex (χ_1).

Remark 4.3 (Physical space representation of the nuclear force). The propagation of the nuclear force in physical space $\mathbb{N}^*$ involves two mechanisms simultaneously: baryons emerge as ζ poles in n -space, while χ_1 vertices propagate transitions through β -space.

5 Response Trace Formula for Coupling Constants

§4 established the Green response structure of the five forces. The type of force is determined by the Green kernel, but how strong the force is—the coupling constant—requires another set of data. In the Standard Model, the five coupling constants (α_{EM} , α_W , α_s , G_N , α_{nuclear}) are independent experimental inputs with no theory relating them to one another. In the BC framework, these five constants have been separately derived in the series:

- [14] gives the three gauge couplings: $\alpha_i = |R_{k_i}|^2/N_i$, where vertex amplitude $R_k = L(\frac{1}{2}, \chi_k)/\zeta(\frac{1}{2})$, and channel number N_i is determined by the algebraic properties of the character;
- [15] gives the gravitational coupling: $G(N) = 1/(E(N)^2 N)$, where $E(N)$ is the radial state-counting formula;
- [16] gives the nuclear coupling: $\alpha_{\text{nuclear}} = |R_1|^2 \times |\text{Aff}(\mathbb{F}_7)|_{\text{eff}}$.

These five coupling constants share the same formula structure $\alpha_X = |A_X|^2 \eta_X$. This section, starting from the angular/radial decomposition of §4.1, elucidates the unified origin of vertex amplitude $|A_X|^2$ and channel statistics η_X .

5.1 Angular/Radial Decomposition of Coupling Constants

From the tensor product structure of §2.3, coupling constants decompose along angular/radial directions just as Green kernels do. All coupling constants are given by a unified formula [12]:

Theorem 5.1 (Response trace formula).

$$\alpha_X = |A_X|^2 \eta_X, \tag{64}$$

where $|A_X|^2$ is the Born probability of the vertex amplitude, and η_X is the statistical weight of internal channels.

The physical meanings of $|A_X|^2$ and η_X differ according to the angular/radial direction.

Angular (gauge forces). Gauge-force coupling proceeds via particle exchange: a particle emits a signal in the character χ_k direction (amplitude A_1), propagates through N_i discrete channels, and is absorbed by another particle (amplitude A_3). Emission and absorption are symmetric ($A_1 = A_3$), and the Born rule gives $|A_X|^2 = |R_k|^2$. The N_i equivalent channels are indistinguishable, and probabilities are averaged over channels (dilution): $\eta_i = 1/N_i$; the more channels, the weaker the coupling.

Radial (gravity). Gravity does not propagate through particle exchange (§4.1); there is no emission or absorption. $|A_G|^2 = C^*/\log^2 N_G$ is given by the discrete curvature constant C^* of the radial Hamiltonian $H = \log n$, measuring the strength of the radial potential. $\eta_G = 1/N_G$ is the uniform dilution over N_G background states. The extreme weakness of gravity is due to the enormous dilution by $N_G \sim 10^{35}$ background states.

Angular×Radial (nuclear force). The vertex amplitude of the nuclear force $|A_N|^2 = |R_1|^2$ comes from the angular direction (χ_1 sector), but the channel statistics $\eta_N = |\text{Aff}(\mathbb{F}_7)|_{\text{eff}} = 42/\tau_L$ comes from the inclusive summation of the radial affine group. In contrast to gauge forces where $\eta = 1/N$ (averaging, dilution), the nuclear force η sums over internal degrees of freedom (enhancement)—this is the structural reason why the nuclear coupling ($\alpha_N \sim 14$) is far stronger than gauge couplings ($\alpha \sim 10^{-2}$).

The detailed derivation of each factor is developed below.

Factor one: vertex amplitude $|A_X|^2$. $|A_X|^2$ is the **spectral weight** of the Dirac operator in the corresponding direction, split into two cases along the angular/radial decomposition.

Angular (gauge forces and nuclear force). In the BC spectral triple $(\ell^2(\mathbb{N}^*), D, \gamma)$, the character-twisted spectral function is [14]

$$\text{Tr}(U_{\chi_k} |D|^{-s}) = L(s/2, \chi_k). \quad (65)$$

The natural order of the spectral action is $s = 1$ (KMS phase transition point); normalizing by the total spectral function $\zeta(1/2)$ gives the vertex amplitude

$$R_k = \frac{L(1/2, \chi_k)}{\zeta(1/2)}, \quad (66)$$

and the combined amplitude $|A_X|^2 = |R_k|^2$ (emission×absorption, Born rule). $|R_k|^2$ measures the weight of character χ_k in the total spectral function. The three gauge forces and the nuclear force share this structure, differing only in the character χ_k .

Radial (gravity). The vertex amplitude of gravity comes from the discrete curvature constant of the radial Hamiltonian $H = \log n$ (§4.3.1). Define the discrete curvature constant of H on $\mathbb{N}^*$:

$$C^* = 2 \sum_{n=1}^{\infty} (\nabla H(n))^2 = 2 \sum_{n=1}^{\infty} [\log(n+1) - \log n]^2 = 1.954378, \quad (67)$$

where $\nabla H(n) = H(n+1) - H(n) = \log(1 + 1/n)$ is the discrete gradient of H . C^* is the discrete curvature constant of the radial Hamiltonian, structurally symmetric with the angular $|R_k|^2$ (squared character spectral weight). The gravitational amplitude is

$$|A_G|^2 = \frac{C^*}{\log^2 N_G}, \quad (68)$$

where N_G is the effective number of background states, and $\log^2 N_G$ is the normalization factor, corresponding to $|\zeta(1/2)|^2$ in the angular direction.

Mixed (nuclear force). The vertex amplitude of the nuclear force $|A_N|^2 = |R_1|^2$ comes from the angular direction (χ_1 sector, quark character), sharing the same amplitude as the strong force. However, its response trace $\eta_N = |\text{Aff}(\mathbb{F}_7)|_{\text{eff}}$ comes from the inclusive

summation of the radial affine group. $|A|^2$ is determined by the angular direction, η by the radial direction—the two factors of the nuclear force belong to different directions, consistent with the structure of the nuclear force equations in §4.3.2 involving both angular and radial directions simultaneously. This angular–radial coupling is the essential feature distinguishing the nuclear force from gauge forces (purely angular) and gravity (purely radial).

Factor two: Born response trace η_X . η_X encodes the statistical treatment of internal channels, likewise classified by angular/radial/mixed.

Angular (gauge forces). The N_i angular discrete paths are indistinguishable; the density matrix is $\rho = I/N_i$, and the response trace is the path averaging:

$$\eta_i = \text{Tr}(\rho^2) = 1/N_i. \quad (69)$$

The channel number N_i is uniquely determined by the algebraic properties of the character [14]:

$$N_i = |\ker \chi_i| \cdot d_2 \cdot c_i, \quad (70)$$

where $|\ker \chi_i| = \varphi(p)/\text{ord}(\chi_i)$ (indistinguishable group elements in the character kernel), $d_2 = (p-1)/2 = 3$ (pairing channels of the involution ι), $c_i = 1 + [\chi_i = \bar{\chi}_i]$ (reality doubling for self-conjugate characters). These three factors encode three types of indistinguishability, all intrinsically determined by the algebraic structure of $(\mathbb{Z}/7\mathbb{Z})^*$. More paths means greater dilution and weaker coupling.

Radial (gravity). The response trace for gravity is the uniform averaging over N_G radial background states:

$$\eta_G = 1/N_G, \quad N_G = N_{\text{Pl}}^{\text{eff}} = 7^{41} \tau_L, \quad (71)$$

where 7^{41} comes from the p -adic valuation layer [15], and τ_L is the Loewy non-semisimple correction. $|\text{Aff}(\mathbb{F}_7)| = 42 \equiv 0 \pmod{7}$ makes the group algebra $\mathbb{F}_7[\text{Aff}(\mathbb{F}_7)]$ non-semisimple (Maschke’s theorem fails); the projective cover of the trivial module S_0 has Loewy chain $P_0 : S_0 \rightarrow S_1 \rightarrow \cdots \rightarrow S_5 \rightarrow S_0$, and the product of node factors is [15, 16]

$$\tau_L = \prod_{j=1}^5 T_j = \frac{3}{2|\zeta(1/2)|} = 1.0271. \quad (72)$$

The same τ_L simultaneously corrects gravity and the nuclear force in a direction-dual manner:

- **Gravity:** τ_L enlarges the effective state count $N_{\text{Pl}}^{\text{eff}} = 7^{41} \tau_L$ (increasing dilution);
- **Nuclear force:** τ_L reduces the effective affine dimension $|\text{Aff}|_{\text{eff}} = 42/\tau_L$ (decreasing enhancement).

The gravitational state-count correction and the nuclear response correction are dual manifestations of the same non-semisimple structure $\mathbb{F}_7[\text{Aff}(\mathbb{F}_7)]$. The averaging over $N_G \sim 10^{35}$ background states is the direct cause of the extreme weakness of gravity.

Mixed (nuclear force). The internal affine states $\text{Aff}(\mathbb{F}_7)$ of the nuclear force are not externally observed; probabilities are summed inclusively over all final states:

$$\eta_N = |\text{Aff}(\mathbb{F}_7)|_{\text{eff}} = 42/\tau_L = 40.89, \quad (73)$$

where $|\text{Aff}(\mathbb{F}_7)| = 7 \times 6 = 42$ is the order of the affine group, and τ_L is the Loewy correction [16]. In contrast to gauge forces and gravity where $\eta = 1/N$ (averaging,

dilution), the nuclear force η *sums* over internal degrees of freedom (enhancement); this is the structural reason why the nuclear coupling is far stronger than gauge couplings.

The four-order-of-magnitude span from $\alpha_{\text{EM}} \sim 10^{-2}$ to $\alpha_N \sim 14$ is entirely explained by this statistical dichotomy of averaging (dilution) versus summation (enhancement). Gauge forces, gravity, and the nuclear force all share the same Loewy factor $\tau_L = 3/(2|\zeta(1/2)|) = 1.027$ (§??).

Unified spectral picture. The complete classification of the two factors along angular/radial directions:

Force	$ A_X ^2$ origin	η_X origin	Statistical method
Gauge forces	Angular ($ R_k ^2$, character spectral weight)	Angular ($1/N_i$, discrete paths)	Averaging
Gravity	Radial ($C^*/\log^2 N$, discrete curvature constant)	Radial $\times$ Angular ($1/N_G$, τ_L correction)	Averaging
Nuclear force	Angular ($ R_1 ^2$, χ_1 sector)	Angular $\times$ Radial ($ \text{Aff} _{\text{eff}}$, τ_L correction)	Summation

Only gauge forces are purely angular (propagation between non-trivial characters). Both gravity and the nuclear force pass through the vacuum direction χ_0 , and χ_0 itself is the intersection of angular (trivial character) and radial (ζ pole)—any force passing through χ_0 naturally involves both directions. The Loewy factor τ_L appears simultaneously in the N_G of gravity and the $|\text{Aff}|_{\text{eff}}$ of the nuclear force, precisely because τ_L arises from the non-semisimple structure of $\text{Aff}(\mathbb{F}_7)$ acting on the angular character space—it is the algebraic marker of angular–radial coupling. Both $|A_X|^2$ and η_X are spectral data of the arithmetic Dirac operator D , entirely determined by arithmetic functions of $p = 7$, with no adjustable parameters.

5.2 Summary Table of Five Forces

Force	$ A_X ^2$	η_X	α_X^{-1}	Experiment	Deviation	Source
Strong force	$ R_1 ^2 = 0.345$	$1/N_s = 1/3$	8.49	8.5	−0.1%	XII
Weak force	$ R_3 ^2 = 0.616$	$1/N_W = 1/18$	29.60	30	−1.3%	XII
Electromagnetism	$ R_2 ^2 = 0.048$	$1/N_{\text{EM}}^{\text{eff}} = 1/6.52$	137.04	137.04	< 0.001%	XII
Gravity	$C^*/\log^2 N_G$	$1/N_G$	1.49×10^{38}	1.50×10^{38}	< 1%	V
Nuclear force	$ R_1 ^2 = 0.345$	$42/\tau_L = 40.89$	1/14.11	1/14.0	+0.8%	VIII

Five rows are different evaluations of the same formula (64). The vertex amplitude $|A_X|^2$ is determined by L -functions (angular) or the discrete curvature constant C^* (radial), with no adjustable parameters. The response trace η_X is determined by angular channel numbers, radial state counts, or the order of the $\text{Aff}(\mathbb{F}_7)$ group; for both gravity and the nuclear force, η contains the Loewy factor τ_L (the angular–radial coupling marker). Gauge forces and gravity use path averaging $\eta = 1/N$; the nuclear force uses inclusive summation $\eta = |\text{Aff}|_{\text{eff}}$. The numerical values for the strong and weak forces include virtual process corrections [14]; the complete derivation of each row is given in the corresponding companion paper.

6 Spectral Completeness

§3–§5 have successively presented the physical content of BC spectral dynamics: matter (§3), forces (§4), and coupling constants (§5). A natural question is: do these results

exhaust the full content of the spectrum? A natural question is: do these results exhaust the entire content of the spectrum? In other words, does the spectrum contain components not yet corresponding to physics, or do there exist particles and forces in physics not yet covered by the spectrum? This section proves the answer is no: the spectrum of the arithmetic Dirac operator D yields precisely all particles and interactions of the Standard Model, no more and no less.

6.1 Spectral Finiteness

The reason that the number of particle and force species in BC spectral dynamics is fixed lies in the fact that the spectrum of the Dirac operator D is completely determined by two sets of finite data—angularly there are only 6 characters, and radially there is only 1 pole. These two finitenesses respectively constrain the upper bound on particle species and the number of emergent phase transitions.

Theorem 6.1 (Spectral Finiteness).

(1) Angular finiteness.

- $(\mathbb{Z}/7\mathbb{Z})^*$ is a finite cyclic group of order $p - 1 = 6$; the character group $\widehat{(\mathbb{Z}/7\mathbb{Z})^*} = \{\chi_0, \chi_1, \dots, \chi_5\}$ has exactly 6 elements (Peter–Weyl).
- The CRT decomposition $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ classifies the 6 characters by order into 4 classes: order 1 (χ_0), order 2 (χ_3), order 3 (χ_2, χ_4), order 6 (χ_1, χ_5).
- The CRT coordinate differences in the adjoint representation $\text{End}(\mathbb{C}[(\mathbb{Z}/7\mathbb{Z})^*])$ have exactly $|\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}| = 6$ types, divided by the angular Laplacian eigenvalues into 4 sectors: $\lambda = 0$ (χ_0), $\lambda = 4$ (χ_3), $\lambda = 3$ ($\chi_2 \oplus \chi_4$), $\lambda = 1$ ($\chi_1 \oplus \chi_5$).
- The 6 character directions exhaust all possibilities for angular positions of particles: χ_0 is the vacuum reference direction, and baryons project back to χ_0 via ε^{abc} ; the 5 nontrivial directions χ_1 – χ_5 label the three generations of fermions and the three gauge force sectors through the CRT decomposition.
- The finiteness of $\varphi(7) = 6$ excludes a fourth generation of fermions and a fourth gauge force.

(2) Radial uniqueness.

- The Riemann ζ function has exactly one simple pole at $\beta = 1$ in $\text{Re}(\beta) > 0$ (Euler 1740).
- For all nontrivial characters, $L(\beta, \chi_j)$ is regular and finite at $\beta = 1$ (Dirichlet 1837).
- The unique pole of $\zeta(\beta)$ gives the unique divergence point of the density of states, which is both the critical temperature for baryon emergence (§4.3) and the phase transition point of χ_0 condensation (§3.2).
- The finiteness of $L(1, \chi_j)$ implies that no emergence occurs in nontrivial directions—emergence occurs only in the χ_0 direction.
- There is no second pole, which excludes additional phase transitions and additional Higgs sectors.

6.2 Spectral Completeness

The two finiteness conditions of §6.1 constrain the number of particle and force species to the following exact count.

Theorem 6.2 (Spectral Completeness). *In the BC spectral dynamics framework:*

1. There are exactly 12 fermions (three generations, four sectors);

2. There are exactly 5 bosons: gauge carriers γ, g, W, Z + nuclear force carrier π ;
3. There is exactly 1 amplitude mode of χ_0 condensation;
4. There are exactly 3 gauge forces (electromagnetic, weak, strong);
5. There is exactly 1 gravitational force;
6. There is exactly 1 nuclear force.

Total: 12 fermions + 5 bosons + 1 condensation mode = 18 particle states, 5 interactions.
Cannot be increased, cannot be decreased.

Proof. In six steps, each derived from the finiteness of Theorem 6.1.

(1) Fermions = 12.

- $|(\widehat{\mathbb{Z}/7\mathbb{Z}})^*| = 6$. Among these, χ_0 is trivial (vacuum direction) and does not correspond to a fermion.
- The 5 nontrivial characters are divided via the CRT isomorphism $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ into three conjugate pairs/sectors: $\{\chi_3\}$ (order 2, self-conjugate), $\{\chi_2, \chi_4\}$ (order 3, conjugate pair), $\{\chi_1, \chi_5\}$ (order 6, conjugate pair).
- Each sector labels one class of fermions by CRT coordinates: neutrinos (χ_3), charged leptons (χ_2, χ_4), quarks (χ_1, χ_5).
- Including particle–antiparticle and three generations, the total is $4 \times 3 = 12$.
- A fourth generation is impossible: $\varphi(7) = 6$ restricts nontrivial characters to 5; a fourth generation would require $\varphi(p) \geq 12$, i.e. $p \geq 13$, contradicting the minimality condition $6|(p-1)$ for $p = 7$ (§??).

(2) Bosons = 5.

- Gauge carriers: the angular Laplacian Δ_{mult} on the adjoint representation has 4 distinct eigenvalues $\{0, 1, 3, 4\}$.
- The factor-preserving automorphism principle excludes $\lambda = 1$ (see step (4)), and the remaining 3 allowed sectors each correspond to one gauge carrier: $\lambda = 0$ (γ), $\lambda = 4$ (W, Z), $\lambda = 3$ (g), giving 4 gauge bosons in total.
- Nuclear force carrier: the π meson is determined by $\chi_1 \cdot \bar{\chi}_1 = \chi_0$ (color singlet), with spin $J = 0$ ($\chi_0(g^3) = 1$ does not change sign), hence a boson.
- Total: $4 + 1 = 5$ bosons.

(3) χ_0 condensation amplitude mode = 1 (i.e. Higgs).

- χ_0 is the unique trivial element of $(\widehat{\mathbb{Z}/7\mathbb{Z}})^*$, so the vacuum direction is unique.
- The breaking direction is determined by the unique nontrivial real character χ_3 (Theorem 3.10): $\chi_k = \bar{\chi}_k \iff 2k \equiv 0 \pmod{6} \iff k \in \{0, 3\}$; excluding the trivial $k = 0$ leaves only $k = 3$.
- Uniqueness of χ_0 (condensation channel) + uniqueness of χ_3 (breaking direction) $\Rightarrow$ the amplitude mode is unique.
- This mode is the experimentally observed Higgs particle (125 GeV)—not an excitation of a fundamental field, but an oscillation of the χ_0 condensate about the new ground state.

(4) Gauge forces = 3.

- The eigenvalues of Δ_{mult} on the adjoint representation are $\lambda_k = 2 - 2\cos(2\pi k/6)$, and $k = 0, \dots, 5$ gives four distinct values $\{0, 1, 3, 4\}$.
- $\lambda = 1$ corresponds to endomorphisms with CRT coordinate difference $(1, \pm 1)$, which simultaneously change both the $\mathbb{C}^2$ (weak) and $\mathbb{C}^3$ (color) factors, thus not preserving the factor algebras $\mathcal{A}_2$ or $\mathcal{A}_3$.
- The factor-preserving automorphism principle excludes this sector [14].

- The remaining three sectors correspond precisely to the three gauge forces of the Standard Model: $\lambda = 0$, $U(1)$ (electromagnetic); $\lambda = 4$, $SU(2)$ (weak); $\lambda = 3$, $SU(3)$ (color).
- $\cos(2\pi k/6)$ takes only the values $\{1, 1/2, -1/2, -1\}$, which are completely exhausted—no fifth gauge force exists.

(5) Gravity = 1.

- Gravity corresponds to the zero mode $\lambda_0 = 0$ of Δ_{mult} (§4.3.1).
- $\chi_0(g) = 1$ for all $g \Rightarrow \Delta_{\text{mult}}\chi_0 = (S + S^{-1} - 2I)\chi_0 = 0$.
- The trivial character is unique in the character group of a finite cyclic group, so $\dim \ker \Delta_{\text{mult}} = 1$.
- There is no second zero mode, hence no second gravitational force.

(6) Nuclear force = 1.

- The nuclear force is given by χ_1 vertex transitions between baryons (§4.3.2): the emergent states (baryons) reside in the χ_0 direction, and transitions proceed via χ_1 (the color singlet combination $\chi_0 \rightarrow \chi_1 \rightarrow \chi_0$).
- $\zeta(\beta)$ has exactly one pole at $\beta = 1$ for $\beta > 0$ (Theorem 6.1(2)), so emergence occurs only at $\beta = 1$, and the transition channel between emergent states is unique.
- There is no second emergence mechanism, hence no second nuclear force.

□

The above five forces and the matter they act upon are organized by direction, Green kernel, carrier, and constraint as follows:

	Direction	Green kernel	Carrier	Source matter	Propagation	Constraint
EM force	Angular ($\chi_0, \lambda=0$)	$L(\beta, \chi_2)$	γ	$e, \mu, \tau, u, d, c, s, t, b$	Particle exchange	
Weak force	Angular ($\chi_3, \lambda=4$)	$L(\beta, \chi_3)$	W, Z	All fermions	Particle exchange	
Color force	Angular ($\chi_2 \oplus \chi_4, \lambda=3$)	$L(\beta, \chi_1)$	g	u, d, c, s, t, b	Particle exchange	$P_{\text{mix}}F=0$
Gravity	Radial	$\zeta(\beta)$	(Collective potential)	All	No propagator	Equivalence principle
Nuclear force	Angular×Radial	$L \otimes \zeta$	π	$p, n, \Lambda, \dots$	Linked transition	Color singlet

The 12 fermions are fully covered by the three gauge forces: electromagnetic, weak, and color; baryons are covered by gravity and the nuclear force. The amplitude fluctuation of the χ_0 -direction condensate (Higgs) is a non-fundamental excitation and is not listed in the table above.

7 Conclusion

This paper provides a unified formulation of the common mathematical objects repeatedly used throughout the series as BC spectral dynamics. The tensor product structure of the arithmetic Dirac operator $D = H \otimes I_F + \gamma \otimes D_F$ separates the physics into angular and radial directions: angular character eigenmodes yield the identity and mass of fermions, angular Green kernels $L(\beta, \chi_k)$ propagate gauge forces, the radial ζ pole gives baryon emergence and gravity, and the combined angular–radial structure gives the nuclear force.

The coupling strengths of the five forces are given by the unified response trace formula $\alpha_X = |A_X|^2 \eta_X$. The vertex amplitude $|A_X|^2$ is determined by L -functions (angular) or the curvature constant C^* (radial), and the channel statistics η_X is determined by character algebra (angular) or the $\text{Aff}(\mathbb{F}_7)$ group structure (radial). Gauge forces and gravity correspond to path averaging ($\eta = 1/N$, dilution), while the nuclear force corresponds to inclusive summation ($\eta = |\text{Aff}|_{\text{eff}}$, enhancement).

The spectral completeness theorem (Theorem 6.2) deduces from the finiteness of $\varphi(7) = 6$ and the unique pole of $\zeta(\beta)$ the exact count of 17 particles and 5 forces, which can neither be increased nor decreased. The factor-preserving principle excludes the $\lambda = 1$ weak-color mixing sector, thereby ruling out leptoquarks, proton decay, a fourth generation of fermions, and additional Higgs sectors.

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